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Prof. Dr. Lars Lindner
Sarcoma / Hyperthermia / Drug Delivery Working Group
Medizinische Klinik III (Onkologie)
LMU University Hospital Munich

Re: PhD / Postdoctoral Position — In Vivo Cancer Therapeutics

Dear Prof. Lindner,

I am applying for the PhD / Postdoctoral position in in vivo cancer therapeutics. My doctoral research was in Computational Lipidomics at TUM, where I developed mass spectrometry pipelines specifically for lipid systems characterisation. That background connects directly to the core analytical problem of your group's work: thermosensitive liposomal drug delivery depends on phospholipid phase transitions that are precisely the events that lipidomics mass spectrometry measures, identifies, and can now predict from first principles.

Lipidomics and liposomal drug delivery. At TUM / Bayerisches Forschungsinstitut I built the complete data pipeline for lipid species analysis: peak detection and deconvolution, ion annotation against lipid databases (LIPID MAPS, LipidBlast), multi-omics statistical modelling, and federated learning architectures for distributed clinical data. Separately, at Fraunhofer Institut IGB Stuttgart, I applied dynamic programming and linear programming methods to the production optimisation of *therapeutic lipids* — the same lipid classes that constitute thermosensitive liposomal formulations. The Lavoisier framework I built for mass spectrometry extends this: the virtual instrument at lavoisier.vercel.app/experiment runs forward simulation of mass spectrometry experiments in the browser, and the underlying framework processes mzML datasets exceeding 100 GB, achieving 98.9% feature extraction accuracy against the NIST reference library. The phase transition of DPPC/DSPC-based thermosensitive liposomes from gel to liquid crystalline phase at approximately 42 °C is observable by mass spectrometry as a characteristic shift in the phospholipid ion population — a measurement class the pipeline was built to characterise.

Why sarcoma and pancreatic cancer are immunologically cold: a first-principles account. My manuscript *On the Thermodynamic Consequences of Categorical Completion on Biological Systems* models adaptive immune selectivity as a categorical filter cascade: MHC molecules present peptides preferentially from proteins with high categorical richness R — quantifying isoform diversity, conformational entropy, and post-translational modification landscape. The correlation is $r = 0.73$, $p < 10^{-12}$ across 2,847 IEDB peptides; adding the R term to NetMHCpan raises AUC from 0.68 to 0.81. Sarcoma is paradigmatically collagen-driven: the dense ECM of soft-tissue sarcomas is dominated by collagen isoforms with $R \approx 10^{5.5}$, categorically the highest richness in the human proteome, which

predicts them as strong immune targets — yet sarcomas are immunologically cold. The model resolves this: high- R target availability does not guarantee immune activation when the tumour microenvironment suppresses the MHC-I presentation pathway. Pancreatic cancer presents the same structure amplified by desmoplastic stroma. The combination of hyperthermia (which disrupts stromal architecture and temporarily restores MHC-I surface density) with liposomal drug delivery and immunotherapy is precisely the intervention the framework predicts will convert these tumours from cold to hot. I can derive specific predictions about which peptide populations will expand following hyperthermia treatment, and which drug targets will have the lowest off-target burden — predictions testable in your group's in vivo models.

Tumour microenvironment as a trajectory completion problem. The Syndrome framework computes cellular disease state as a vector $\mathbf{D} \in [0, 1]^8$ across eight oscillator classes. For the hyperthermia / liposomal drug delivery context, the directly relevant components are the membrane potential oscillator (D_M , responding to thermal stress-induced bilayer perturbation), the calcium oscillator (D_{Ca} , which governs the heat-shock response and drives immunogenic cell death signalling), and the ATP oscillator (D_{ATP} , depleted under the metabolic demands of combined hyperthermia and cytotoxic drug exposure). The framework computes the trajectory from tumour cell state to the target state (immunogenic, drug-sensitive, cleared) in $\mathcal{O}(N \log N)$, and the sparse ℓ_1 linear programme selects the minimum-intervention combination — which of hyperthermia dose, liposomal drug concentration, and immunotherapy agent — needed to complete the transition. This is exactly the optimisation problem your multimodal therapy programme needs to solve.

Computational infrastructure for the experimental programme. The tools I have deployed address each analytical dimension the programme requires. vivid-symbolism.vercel.app/search provides sequence homology search underpinning tumour neoantigen identification. shakespear-nine.vercel.app/instrument provides protein folding, trajectory, and dynamics analysis applicable to the oncoprotein targets released by hyperthermia-mediated cell lysis. The Borgia framework derives molecular spectroscopic properties from bounded phase space geometry — the Triple Equivalence Theorem equates oscillatory dynamics, categorical state enumeration, and partition operations, making it possible to predict the DPPC and DSPC vibrational spectra and phase transition temperatures from structural geometry alone, without parameter fitting against empirical data. The cheminformatics instrument at honjo-masamune.vercel.app/tools implements this directly: zero-parameter molecular spectroscopy including docking, 3D structure, and harmonic network analysis, all computed in the browser.

Background. My MSc in Pharmaceutical Biotechnology (HAW Hamburg) covered drug delivery, formulation science, and pharmacokinetics. My BSc in Biochemistry and Cell Biology (Jacobs University Bremen) included standard cell culture and molecular biology laboratory training. My MSc in Computational Biology (Universität Tübingen) and doctoral research in Computational Lipidomics at TUM gave me the analytical and statistical depth behind the work described above. I am not proposing to arrive as an animal model specialist — that is the training this position provides. I am proposing that a candidate who can derive from first principles why sarcoma and pancreatic cancer are immunologically cold, predict which hyperthermia-driven state transitions will produce immunogenic cell death, and analyse the resulting lipidomic and proteomic data at scale will contribute substantively to this programme from the first day.

I am legally resident in Germany with full work authorisation and am available from October 2026. English is native; German is conversational.

Yours sincerely,

Kundai Farai Sachikonye